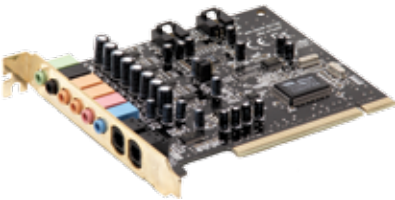


turn is connected to speaker cabinets. The number of channels on a mixer depends on where it is being used. A nightclub DJ might need only two channels, whereas a live rock concert or a recording studio might need as many as 72 channels.

It would be wrong to assume that all they can do is mix. They can do a lot more for the sound like pan control, filtering and equalisation, effects and more.

Sound Cards

Almost all of us know extensively about graphics cards and their benefits. Just like graphics cards boost video performance on your system, sound cards do the same to audio performance. Essentially, a sound card is a digital to analog converter (DAC), which converts recorded digital audio into an analog output and sends it to amplifiers, headphones or external devices using standard connectors.



A sound card. This one is from Turtle beach

Advanced cards include powerful processing capabilities that allow higher data rates and intensive audio operations like producing synthesised sounds for real time music generations and sound effects without burdening the CPU. Multi-channel

processing allows simultaneous generation of different sounds with different volume and pitch, along with effects like filtering and distortion. This approach to music generation is being used by manufacturers to produce low cost cards.

Most cards accept external input from sources that have higher voltage levels than a microphone. This signal is digitised and stored on the hard drive for future processing.

A significant characteristic in sound cards is polyphony, i.e processing and output of multiple independent sounds and voices simultaneously. The multi-channels mentioned earlier can be seen as the number of audio outputs which in turn correspond to the speaker or output configurations like 2.0 (stereo), 2.1 (stereo with subwoofer) 5.1(surround) and more. For some years, most sound cards have multiple frequency modulated sound channels (typically 9 to 16) but sound cards aren't usually used to their full potential. Typical digital sound samples are played using one or two